

Does insulin sensitivity vary with both glucose and total daily dose, and is there a clear relationship between the three?

Testing both dynamic ISF equations against 1,684 people in seven public study archives

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What this is about

Somewhere in your pump settings is a number that says how far one unit of insulin drops your glucose. Most of us arrived at it from the 1800 rule, which divides 1800 by your total daily dose, then adjusted it when corrections kept landing too hard or too soft.

Dynamic ISF replaces that fixed number with a calculated one. Instead of using what you entered, the algorithm works out a fresh sensitivity every few minutes from two things: how much insulin you have been using lately, and where your glucose is right now.

There are two versions of the calculation and they disagree about how much your daily dose should matter. The original, v1, keeps the 1800 rule's shape: if your daily dose doubles, your sensitivity halves. The revised version, v2, squares the dose term, so if your daily dose doubles, your sensitivity drops to a quarter.

That difference is not academic. Take someone on 20 units a day and someone on 60. v1 says the second person's corrections should be three times smaller per unit. v2 says nine times smaller. The two versions cross at around 64 units a day, so which one you are running decides whether the newer maths makes your corrections stronger or weaker than the older one.

Both also reduce your sensitivity when your glucose is high, on the reasoning that insulin works less well when you are running high.

This is an attempt to check all of that against a lot of people.

The short version

Your daily dose does matter, and roughly in the way v1 says. Sensitivity falls as daily dose rises, at an exponent close to what the 1800 rule assumes, though a little shallower. Four separate methods agree on this, including one that was free to find no relationship at all.

Your glucose level also matters, but in the opposite direction to what both equations assume. They make insulin most effective when you are near target and least effective when you are high. What I measured is the reverse. A unit does least near target, which is where your body is actively pushing back against going lower, and slightly more when you are high.

The two do not combine the way either equation combines them. Both multiply a dose term by a glucose term and assume the two are independent. They are not: how your sensitivity changes with glucose depends on how much insulin you use.

And a real part of what the equations are picking up is food rather than sensitivity. In the four hours after a meal you logged, measured sensitivity goes negative, meaning glucose rises while insulin is acting, because carbohydrate is still arriving after the absorption model has finished counting it.

When I asked each version to predict how far glucose would actually fall overnight, a plain static 1800 rule did better than either of them, and v2 did very badly indeed.

What that means if you are running one of these

If you use v1, the dose half of it is close to right and the glucose half is pointed the wrong way but is small enough not to do much harm.

If you use v2, the squared dose term is not supported by anything I can measure here. It was not the best predictor for a single person out of 1,626, and it fails worst for people on small daily doses. For a young child on ten units a day, squaring the dose gives a sensitivity of several thousand, which means a correction dose of almost nothing.

If you use neither, the 1800 rule you probably already have is a reasonable description of what these archives show, and it beat every dynamic form I tested at predicting the overnight fall.

None of this is a criticism of the people who built these equations. They were working from the same rule of thumb most of us set our pumps from, and one of the findings here is that the rule holds up well.

Where the data came from

Seven public study archives, about 1,700 people, ages 2 to 82, daily doses from 7 to 107 units.

Study	What people were using	People	Food logged	Median days	Median dose
Loop	Do-it-yourself Loop, fixed sensitivity	842	yes	397	38.6 U
REPLACE-BG	Pump and sensor, nothing automating	196	yes	246	41.9 U
DCLP3	Control-IQ, adults and teenagers	112	no	183	50.0 U
DCLP5	Control-IQ, ages 6 to 13	100	no	203	37.4 U
PEDAP	Control-IQ, ages 2 to 5	98	no	273	13.6 U
IOBP2	Bionic pancreas	336	no	93	49.0 U

That gives 708,106 usable overnight stretches from 1,659 people, each contributing at least forty.

Two of those rows matter more than the others.

REPLACE-BG ran in 2015 on a pump and a sensor with no algorithm in between. That makes it the only group where the insulin someone received was not chosen by a machine watching their glucose, which turns out to matter a great deal.

Loop and REPLACE-BG are also the only groups where people recorded what they ate. That makes them the only two where I can genuinely exclude a meal rather than guess at one.

How I worked out what a unit did

None of these archives records an insulin on board figure, or what the algorithm predicted, or in most cases even a sensitivity setting. All of it has to be rebuilt from the basal and boluses that were delivered.

The approach is the one you would sketch on paper. Take a stretch of time overnight. Note how far glucose fell. Work out how much insulin was actually acting across that stretch, adding up every dose that contributed, using a six hour insulin curve. Divide one by the other and you have milligrams per decilitre per unit, which is what a sensitivity factor is.

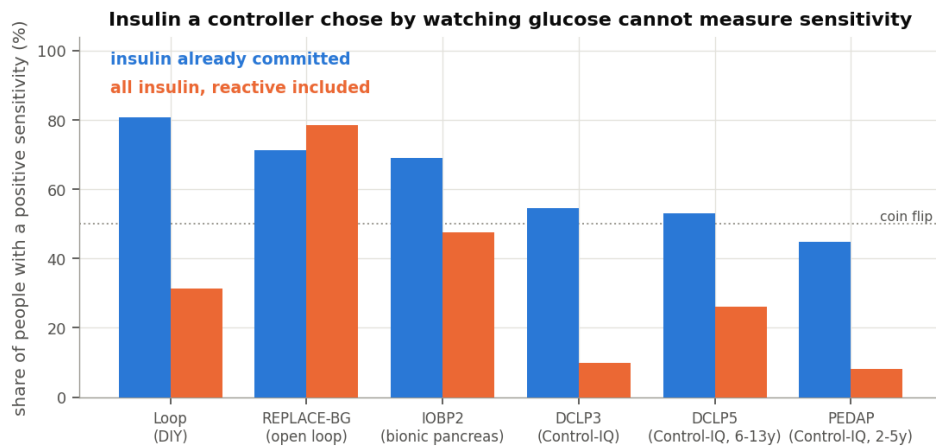
Two details in that do most of the work, and both are worth understanding because they are where this kind of analysis usually goes wrong.

Insulin your loop chose cannot measure your sensitivity

If an algorithm is running, it gives you more insulin exactly when your glucose is refusing to come down. So if you compare the fall against all the insulin that acted, you find that more insulin goes with less falling, and the answer comes out negative. That would say insulin puts your glucose up. It does not. You have measured the algorithm's behaviour rather than your body's.

I handle this by separating insulin according to when it was decided. Insulin already in you when a stretch begins was committed before anything in that stretch happened, so it cannot be a reaction to the glucose I am about to measure. Insulin delivered during the stretch can be.

The size of that problem is worth seeing.



The confound, made visible

Reading across the groups, the share of people whose sensitivity comes out positive collapses from 78% to 8% as the system gets more reactive, then recovers once I use only the insulin that was already committed. REPLACE-BG is the exception, because in 2015 nothing was choosing it.

That is why 196 people from a decade ago carry more weight here than their number suggests.

Basal is not free

Over any stretch of time, your glucose change is insulin action minus the glucose your liver is putting out. Basal exists to cancel that output. So when your basal is right, your glucose is flat, and a naive calculation of fall divided by insulin gives zero. That is not a sensitivity of zero, it is a correct basal rate.

When I ran that naive calculation it returned 3 to 9 mg/dL per unit against entered settings of 25 to 60, which is that effect and not a finding. Everything below therefore compares each stretch against what that person's glucose usually does at that time of day, and against how much insulin usually acts then. What survives is the effect of the insulin that was not routine, which is what a correction dose actually is.

A check that the arithmetic is right

Loop wrote down its own insulin on board figure at every bolus. That is the one place in all of these archives where an app's internal state was recorded, so I can rebuild it from the pump data and see whether I get the same answer.

Across 158 people, my reconstruction tracks Loop's own figure at a correlation of 0.927, with a typical difference of a third of a unit. That is the whole dose calculation checked against something other than my own assumptions.

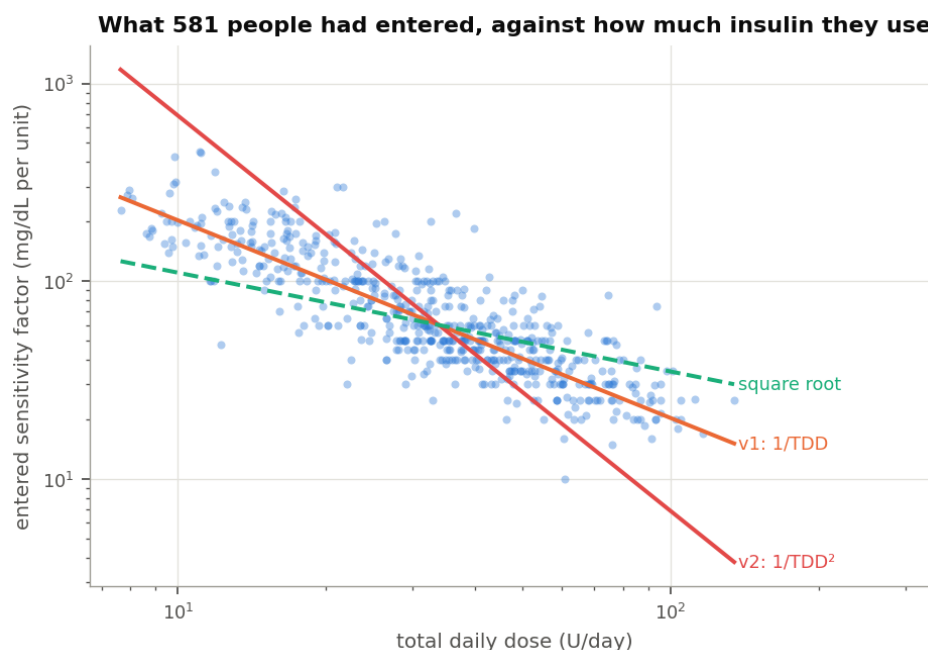
Getting there needed the basal profile, which is in no settings file in the archive, because Loop counts insulin on board net of scheduled basal. It turned out to be recoverable: every temp basal record carries the rate it was overriding.

Question one: does your daily dose change your sensitivity?

Yes, and by roughly the amount the 1800 rule assumes.

What people had entered

Of the 596 people with a sensitivity factor on record, 581 also have enough pump data to work out a daily dose.



Entered sensitivity against daily dose

Plotted against dose on a log scale, the slope is -0.979, with a 95% interval running from -1.030 to -0.934. An exponent of exactly -1 is the 1800 rule, so that interval sits on it and excludes anything much shallower.

Multiply each person's sensitivity by their daily dose and the median comes to 2067. The 1800 rule, evaluated the way v1 evaluates it, gives 2139. If the rule holds, that product should not drift as dose changes, and it does not: the

correlation between them is $+0.024$, which is as close to nothing as you will see in real data. Run the same test on v_2 's squared version and you get $+0.859$, which is what a rule looks like when it does not hold.

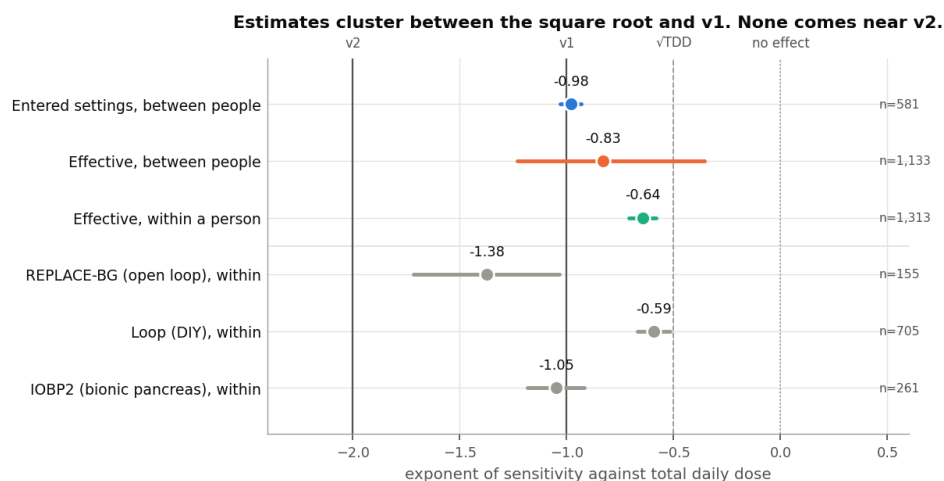
There is a caveat here that needs saying out loud, because it is close to circular. What you have entered is a decision, not a measurement, and you and your team probably reached it through the 1800 rule in the first place. So this is partly a finding about how much grip that rule has on practice.

It is not circular the other way round, though. Nothing in clinical practice pushes anyone towards a squared law, and nothing in what people run looks like one.

Carb ratios follow the same pattern more loosely. Multiplying each person's carb ratio by their daily dose gives a median of 411, so the 500 rule describes practice less tightly than the 1800 rule does.

What people's glucose actually did

Measuring from the glucose response rather than the settings gives a slightly shallower relationship. Between people it comes to -0.83 . Within a single person, which is what these equations actually do several times an hour, it comes to -0.645 across 1,313 people, with a standard error of 0.033 and about three quarters sharing the direction.



Exponents measured by each method

For a sense of what those numbers mean: an exponent of -1 is v_1 , where doubling your dose halves your sensitivity. An exponent of -2 is v_2 , where doubling your dose quarters it. An exponent of -0.65 means doubling your dose multiplies your sensitivity by about 0.64, so a bit more than half.

Both sit between the square root and v_1 . Neither is anywhere near v_2 .

The same question asked at every reading

The measurements above fit a line through each person's data. The more direct thing is to work out a sensitivity at every single glucose reading and put it beside what each equation calculates at that same moment.

At each reading I look back six hours, add up the insulin that acted in that time from every dose that contributed, and compare the glucose change against what that person's glucose usually does over the same six hours at that time of day. Only stretches with no food recorded are used. That gives 1,408,861 readings from 1,004 people.

The level that comes out is too low to read directly, because dividing one noisy number by another pulls the middle of the distribution towards zero. The relationship survives that, and there is a way to check it: I ran the same calculation on the numbers v_1 and v_2 themselves produce, where the answer is known in advance.

What was measured	Exponent recovered	What it should be
Real sensitivity	-0.843 (95% CI -0.970 to -0.720)	unknown, that is the question
v_1 's own output	-1.026	-1
v_2 's own output	-2.085	-2

The method recovers a known answer to within about 0.03. On real data it gives -0.843, which lands on the -0.83 from the completely different method above.

The same calculation shows how far off the two equations are in absolute terms. Across all 1.4 million readings, v_1 sits at about two and a half times the measured value and lands within 30% of it 16% of the time. v_2 sits at five and a half times and lands within 30% of it 9% of the time.

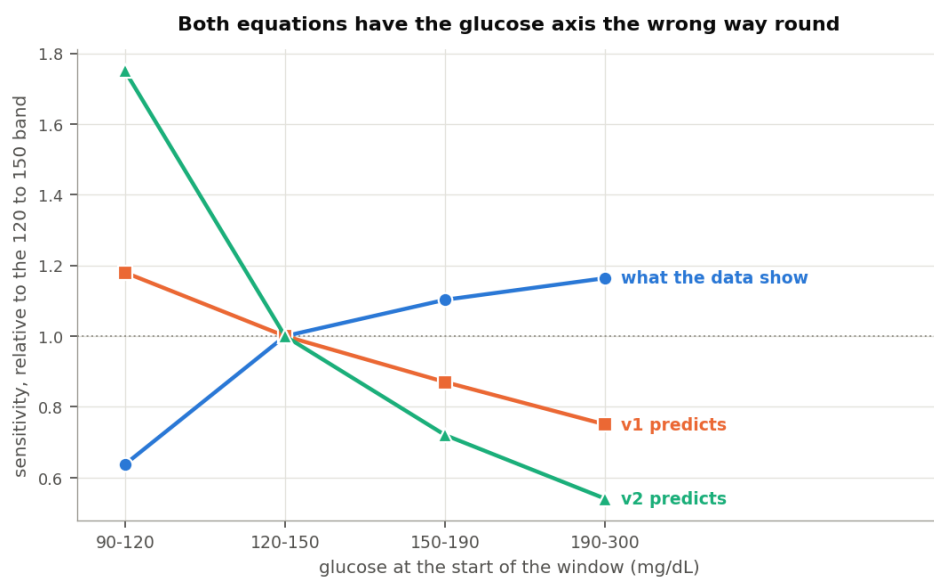
So the dose relationship is real, it is close to what the 1800 rule assumes, and it is nothing like the squared version.

Question two: does your glucose level change your sensitivity?

Yes, and in the opposite direction to the one both equations assume.

This one has a trap in it. Glucose that starts high falls further than glucose that starts low whatever insulin is doing, partly through your kidneys clearing it and partly because things that are high tend to come down. And near target your body pushes back against going lower. So if you let glucose explain the size of the fall, and then ask whether the fall per unit also depends on glucose, you will answer the first question and write down the second.

Every calculation here therefore separates the two, and what carries the claim is only the part that says a unit of insulin itself did more or less.



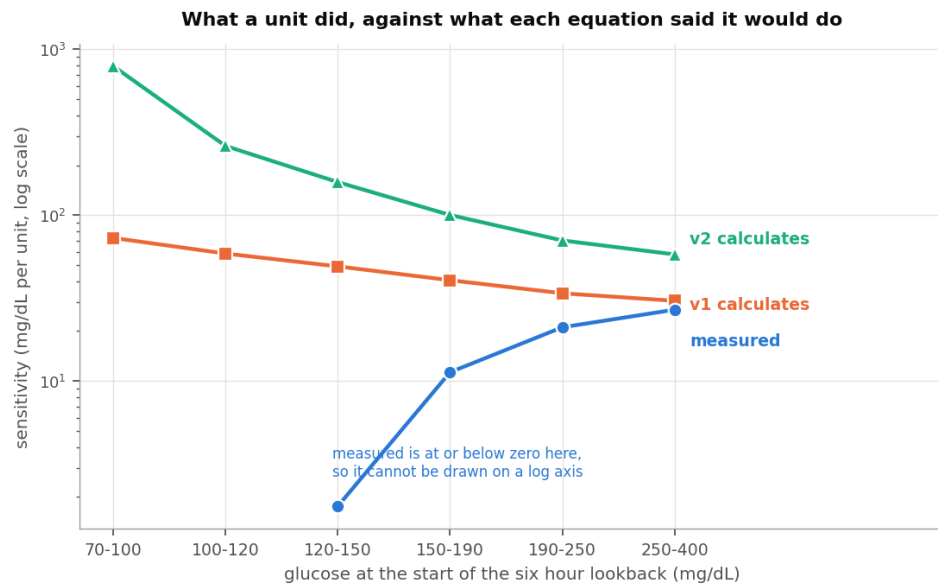
Sensitivity by glucose band against both equations

Where your glucose was	What I measured	v1 assumes	v2 assumes
90 to 120 mg/dL	0.64	1.18	1.75
120 to 150	1.00	1.00	1.00
150 to 190	1.10	0.87	0.72
190 to 300	1.16	0.75	0.54

Those are relative numbers, scaled so the 120 to 150 band is 1.00.

Read the first column downwards. A unit of insulin does about a third less near target than it does in the middle of the range, then a little more as glucose rises. Read the other two columns and they go the other way entirely.

The per-reading calculation says the same thing in real units, and it fits nothing at all.



Measured sensitivity against both equations at the same reading

Where your glucose was	What I measured	v1 calculates	v2 calculates
70 to 100 mg/dL	-0.0	72.9	791.5
100 to 120	-1.5	58.7	261.5
120 to 150	1.8	49.1	158.2
150 to 190	11.3	40.5	100.3
190 to 250	21.1	33.8	70.4
250 to 400	26.9	30.5	58.0

Three different methods return that reversal, and the third one fits no curve to anything.

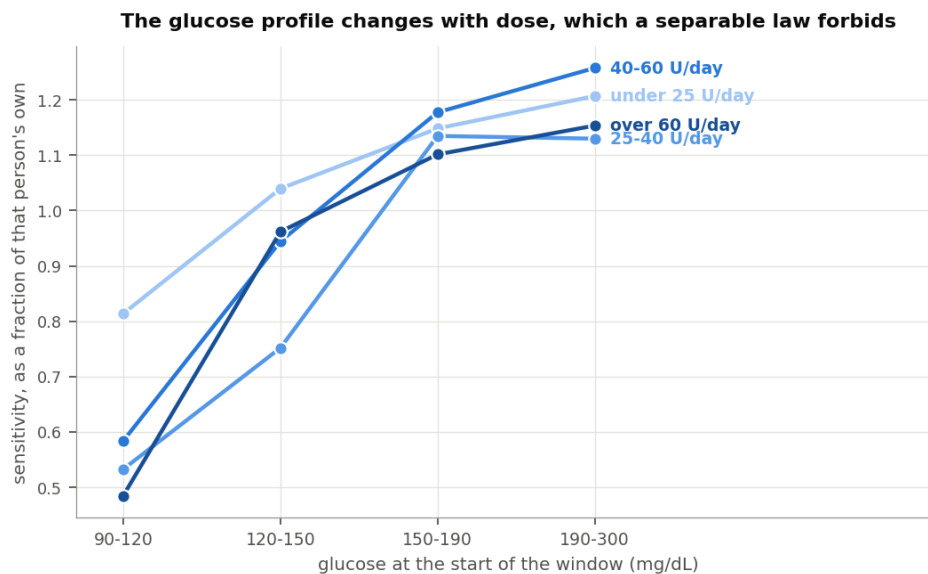
What it means in practice is that both equations correct hardest where insulin is working best, and ease off where it is working least. The near-target end is the one worth thinking about, because that is where your body is defending you against a hypo, and both equations read that defence as high sensitivity and dose into it.

The good news is that the glucose term is small enough not to matter much. Out of sample, six different glucose shapes including both equations' own came within 0.05 mg/dL of each other at predicting the overnight fall, and the flat one won. I had set the bar at 0.5 mg/dL before a glucose term could be said to earn its place. Nothing came within a tenth of that.

Question three: do the two work together the way the equations assume?

No.

Both equations multiply a dose term by a glucose term, which quietly assumes the two do not interact. In plain terms, they assume your glucose profile looks the same whether you take 15 units a day or 80.



Glucose profile at each dose band

Daily dose	90 to 120	120 to 150	150 to 190	190 to 300
Under 25 U	0.81	1.04	1.15	1.21
25 to 40 U	0.53	0.75	1.14	1.13
40 to 60 U	0.58	0.94	1.18	1.26
Over 60 U	0.48	0.96	1.10	1.15

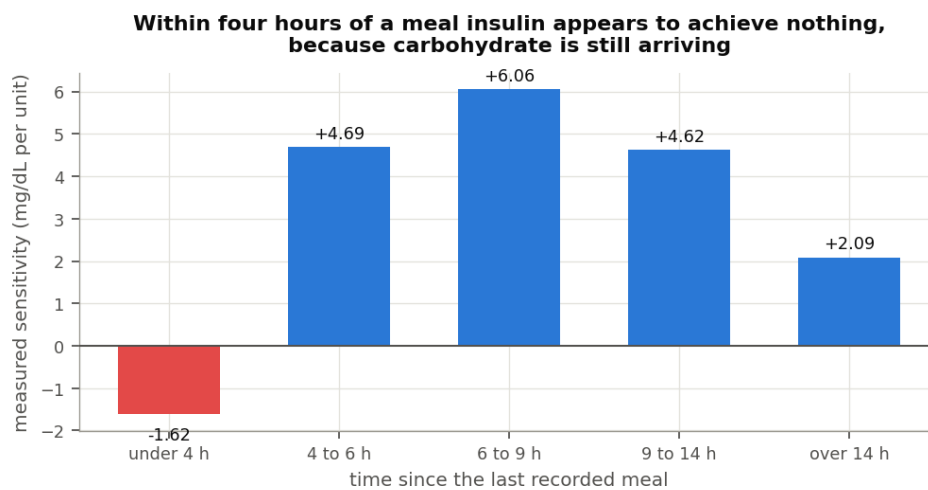
For the equations to be right, those rows would have to be identical. They are not. The dip near target is mild for people on under 25 units a day and around twice as deep for everyone else. Fitting the interaction directly gives -0.444, which is about half the size of the dose term itself and comfortably real.

One thing does behave. Comparing across people, someone's own glucose profile shows no relationship at all with how much insulin they use (a rank correlation of -0.017 across 971 people). It is within a single person that the two axes tangle.

Is any of this really sensitivity, or is it food?

This is the question underneath all of it. Both explanations predict the same thing. If your sensitivity genuinely falls when you are high and when your recent dose is large, the equations describe something real. If instead carbohydrate from a recent meal is still absorbing after the model has finished counting it, then your glucose is high for that reason, insulin appears to be achieving less than it should, and your recent dose is large because you ate. The measurement looks identical either way.

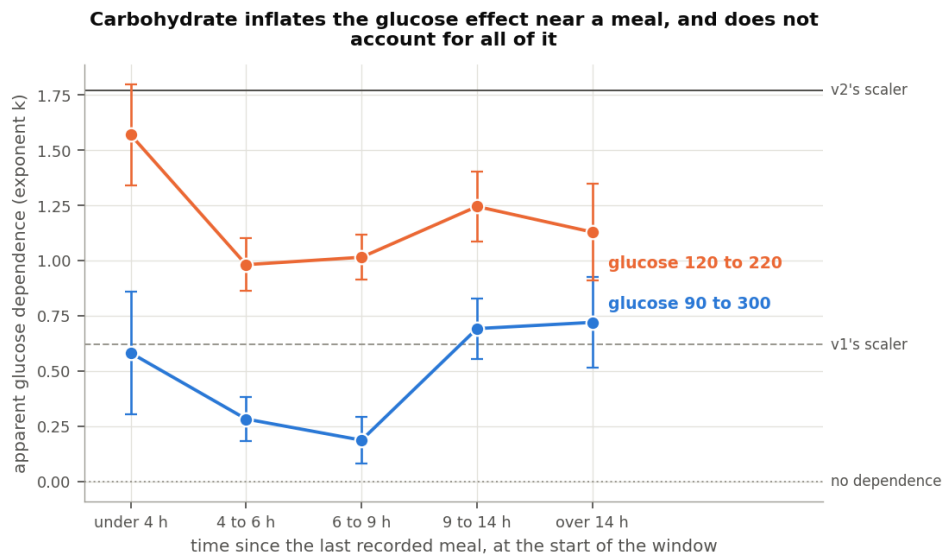
Loop and REPLACE-BG recorded meals, so I can use time since eating to tell them apart.



Measured sensitivity by time since the last meal

Within four hours of a meal you logged, measured sensitivity is -1.62 mg/dL per unit. A negative sensitivity is not a real thing. It says glucose went up while insulin was working, which is what happens when carbohydrate is still arriving from a meal the absorption model has already written off. It recovers to 6.06 by six to nine hours out.

That is the absorption tail, measured directly, and it is not small.



Apparent glucose dependence by time since the last meal

The apparent glucose effect follows the same pattern. Close to a meal it is about three times what it is in clean fasting. So a real part of what the glucose term is responding to is food rather than sensitivity.

It does not go away entirely, though. Even well clear of a meal, some glucose dependence remains, so food does not explain all of it.

The dose relationship is a different story. Holding recent carbohydrate constant, 83% of it survives.

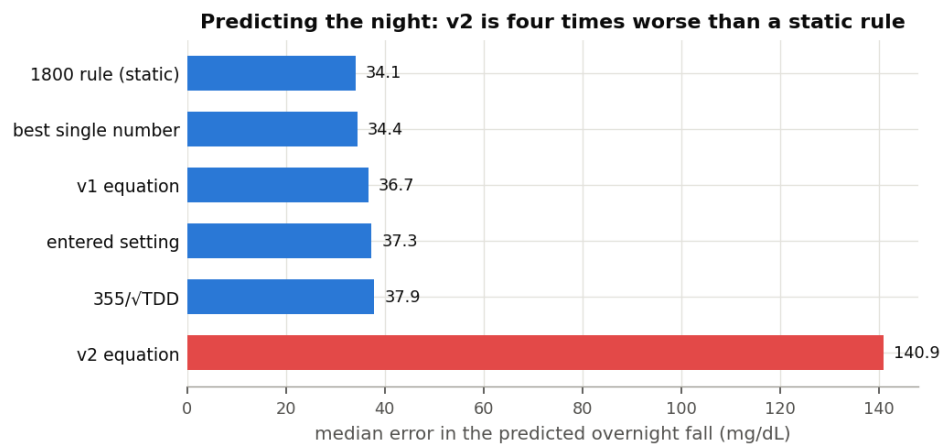
Group	Before	After holding food constant	Kept
Loop	-0.586	-0.500	85%
REPLACE-BG	-1.443	-0.978	68%
Together	-0.648	-0.539	83%

So the dose half is mostly describing something real. The glucose half is substantially food.

Putting it to work: predicting the night

The numbers above are the science. This is what happens if you actually use them.

Each candidate gets its own offset for each person, worked out from the first 70% of that person's nights and scored on the rest. That is generous to the equations rather than harsh on them, because most overnight insulin is basal, and asking a sensitivity factor to account for your liver would be blaming it for something it was never meant to do.



Predicted overnight fall by candidate

What you use	How far out it was, typically
A static 1800 rule	34.1 mg/dL
The best single number for that person	34.4
v1	36.7
Your own entered setting	37.3
355 divided by the square root of your dose	37.9
v2	140.9

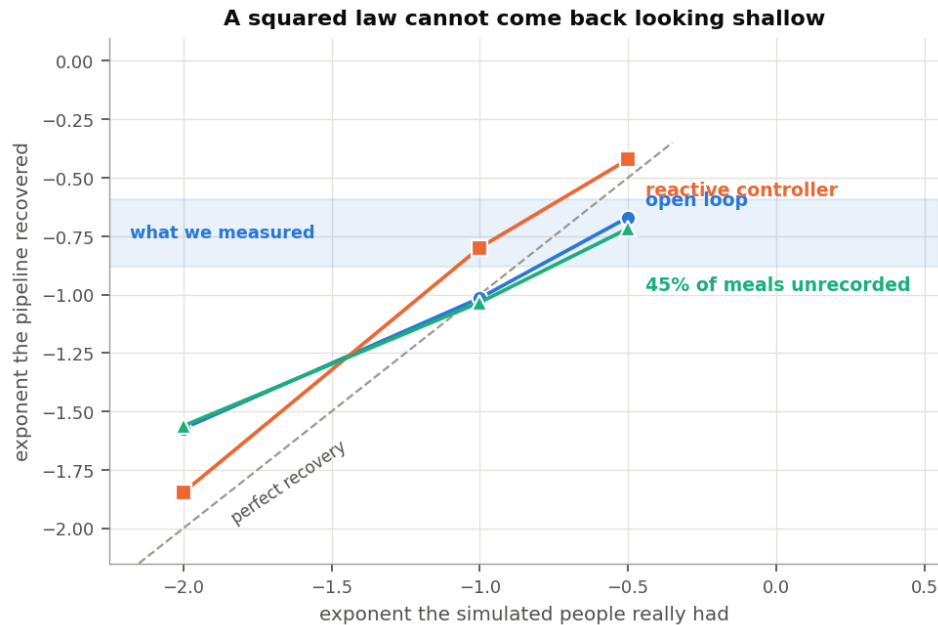
v2 is also biased, not merely imprecise: it overshoots the fall by 41 mg/dL at the median, meaning it consistently expects insulin to do far more than it does.

Out of 1,626 people, the best option was a single well-chosen number for 638, the static 1800 rule for 587, the square root version for 195, v1 for 135, and their own setting for 71. There was nobody it was v2 for.

v2 is worst where it matters most. For people on under twenty units a day its typical error is 322 mg/dL, for the reason given earlier. The error comes down as dose goes up, to 83 mg/dL above 64 units a day, which is the crossover doing what the algebra says it should. Even there it is more than twice as far out as anything else.

How do I know the method itself is not producing this?

Fair question, and the way to answer it is to build people whose sensitivity follows a rule I choose, then run them through the identical machinery and see what comes back.



Recovery of a known relationship

What I built in	What came back
-0.50	-0.50
-1.00	-1.02
-2.00	-1.57

Shallow relationships come back almost exactly. Steep ones come back slightly flattened. The bottom row is the one that matters: a squared law comes back reading between -1.56 and -1.85 under everything I tried, including a run where nearly half of all meals went unrecorded. What I measured was between -0.65 and -0.88. A squared law cannot look like that.

What this does not tell you

It does not give you a number to set. The sensitivities I measure are shrunk by the method, so you cannot read a constant off this work, only the shape of the relationship.

It is all overnight and fasting. The four hours after a meal are measured here only well enough to show sensitivity going negative. Working out what a meal is really doing over those hours would need the carbohydrate absorption model itself under the microscope, which is a different piece of work and probably a more useful one.

It says nothing about sensitivity that moves with the day rather than the dose. Exercise, illness and a change in how you eat all move it over hours and days. None of that is measured here, and none of it is being argued against.

The groups running automated systems are weak evidence on their own. In the Control-IQ studies the share of people with a positive sensitivity is close to a coin toss, and for the youngest group it is worse than one. They back up a direction rather than establishing it.

And two faults in the underlying data had to be repaired along the way. Extended bolus durations arrive in milliseconds for three of the six studies and in seconds for the other two, so a one hour square wave reads as a thousand hours. The database also returns timestamps at microsecond resolution, which divided every basal total by a thousand until I checked the reconstruction against the database's own sums. Anyone else working from these archives will meet both.

Where that leaves the two equations

The dose half of v_1 is close to right, and it is closest in the place it came from: it describes what people actually run their pumps at to within a few per cent. Measured against what glucose does it looks a touch steep, with the supported range somewhere between the square root and v_1 .

Its glucose half is pointed the wrong way, and so is v_2 's. Both make insulin most effective near target when the measurement says it is least effective there. The saving grace is that the term is small enough that it costs little.

For v_2 's squared dose term I could find no support here. What people run contradicts it, what glucose does contradicts it, it was not the best predictor for anyone, and the simulation says a squared law could not have looked like anything I measured. Its worst behaviour is in children and in anyone on a small daily dose, who are the least able to absorb a correction that turns out to be far too small.

I want to be careful about the scope of that. It describes what a large and varied group of people did. It says nothing about anyone's intent or competence, it is one analysis of overnight fasting periods from archives never collected for this purpose, and the equations under examination moved the conversation to where this work could start. Somebody coming at it another way would be welcome.

If you want a dose term, what is here supports something between about -0.65 and -1.0, applied to your own level rather than used to set it. But given that a plain static 1800 rule out-predicted every dynamic version I tested, the question I keep coming back to is not which exponent to use. It is whether sensitivity is the right place to be making this adjustment at all, when the thing most obviously needing attention is what a meal is still doing four hours after you logged it.

Methods

Windows run four hours from starts between 23:00 and 03:00, requiring 80% sensor coverage, a starting glucose between 90 and 300 mg/dL, and either six hours clear of logged carbohydrate or, where none is logged, four hours clear of a bolus large relative to that person's own daily dose. Screening looks only backwards, never at what glucose subsequently did, because dropping the nights where glucose rose removes disproportionately the nights insulin worked least well and inflates the result.

Insulin action comes from convolving the delivery record with an insulin model. Loop subjects use the models LoopKit itself ships, taken from its source rather than approximated. Everyone else uses the oref exponential at six hours with a 75 minute peak.

Per-person fits are ordinary least squares with a heteroskedasticity-robust standard error. Population figures pool the per-person estimates by the method of DerSimonian and Laird, as in the earlier work in this series, so the two can be read side by side. Confidence intervals are percentile bootstrap over people, using the same weights as the estimate they wrap.

The per-reading calculation looks back six hours at every reading, takes both the glucose change and the insulin action as departures from that person's own median for that half hour of the day, and requires the excess insulin action to exceed 0.3 units so that the denominator is not noise.

Separability is tested by fitting insulin action against glucose, against dose, and against the product of the two, and reading the third coefficient. The glucose profile is measured both as a power law and, because the power law turned out to describe it poorly, by interacting insulin action with glucose band indicators, which imposes no shape.

Fitted as a power law rather than as bands, the glucose exponent comes to +0.238 with a standard error of 0.044. That value is not quoted in the body above because it is unstable: restricting to a narrower glucose range moves it to about +1.0, a shift larger than the number itself. A power law cannot describe a step, so fitting one across the profile in the table averages the step into a slope of almost nothing whose sign follows whichever range was chosen. The band table is reported in its place.

An earlier draft of this document reported a glucose exponent of -2.67. That was an indexing defect: the interaction term had been inserted ahead of the insulin coefficient in the design matrix while the coefficient was still read from a fixed position, so the interaction was divided by the starting glucose term. It was found when a second implementation disagreed on the sign. Design matrix columns are now addressed by name and a regression test covers it.

Code is in `inv009/` in the `dynamic-isf-calculations` repository, with 29 tests.

```
psql -d oref -f inv009/ingest/sql/05_settings.sql
python3 inv009/ingest/load_settings.py
python3 -m inv009.build_cache
python3 -m inv009.entered_isf && python3 -m inv009.effective_isf
python3 -m inv009.tdd_axis && python3 -m inv009.glucose_axis
python3 -m inv009.head_to_head && python3 -m inv009.synthetic
python3 -m inv009.loop_model_infer && python3 -m inv009.ml_shap
python3 -m inv009.carb_hypothesis && python3 -m
inv009.joint_surface
python3 -m inv009.pointwise_isf && python3 -m inv009.figures
```

Data attribution

The source of the data is the Loop Study (sponsored by the Jaeb Center for Health Research and funded by the Helmsley Charitable Trust), but the analyses, content and conclusions presented herein are solely the responsibility of the authors and have not been reviewed or approved by the study sponsor.

The source of the data is the Insulin Only Bionic Pancreas Pivotal Trial, but the analyses, content and conclusions presented herein are solely the responsibility of the authors and have not been reviewed or approved by the Bionic Pancreas Research Group or Beta Bionics.

DCLP3, DCLP5, PEDAP and REPLACE-BG are also public datasets from the Jaeb Center for Health Research, whose releases carry no attribution wording of their own. The analyses, content and conclusions here are solely mine and have not been reviewed or approved by any study sponsor.

Dates in these archives are shifted or rebased per participant, so time of day is preserved and calendar date is not. Nothing here depends on calendar date.